Evaluating a Binaural Model Using Real-Time Assessment of Listening Effort in Human Listeners

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Introduction

Estimating speech intelligibility (SI) and listening effort (LE) in real-time can be beneficial for hearing aids to automatically choose the best algorithm at any time. In [1] a concept for such a model was proposed, which could be implemented in real-time. It uses the blind Binaural Speech Intelligibility Model (BSIM) [2] as a front-end, which models the effect of spatial release of masking by enhancing speech accordingly and Listening Effort prediction from Acoustic Parameters (LEAP) [3, 4] as a back-end, which estimates SI and LE. Both stages work blindly in a sense that they don't require any clean source signals or any other auxiliary information.

Usually such models are evaluated using hearing experiments, where participants listen to sentences of speech in noise repeatedly with altering signal-to-noise ratio (SNR), while word recognition rates or the relation of LE depending on SNR is tracked. Continuous model outputs of SI or LE may then be averaged over time for each sentence to be compared to the data. However, to our knowledge so far there exists no concept for evaluating such a model for dynamic acoustic scenes, where parameters may change constantly and SI and LE of participants are assessed in the moment.

In [5] a method was introduced, where the course of signal distortion over time was pre-generated, which was referred to as a profile. Profiles were then assigned to a speech stimulus, which were presented to participants via headphones. Participants were then able to assess perceived speech quality continuously using a hardware slider interface with category labels on it.

In this study a real-time version of the model by [1] is introduced. As of now, the real-time version of the model only predicts LE, which is why only LE is assessed in this study. Additionally, the described approach by [5] was implemented for assessing listening effort to evaluate the real-time model while also investigating the capability to model spatial release of masking. Additionally, response times of participants were tracked to observe how fast a model has to be to adapt to changes.

Methods

Model Overview

The model used in this study uses a new online implementation of the blind BSIM [2] as a front-end, while replacing the LEAP [3, 4] model, which had been used in [1], by an end-to-end trained deep neural network as the back-end for computational efficiency and reduced processing delay.

A block processing scheme was used with blocks of 64 ms with 50% overlap. The blind BSIM applies the Equalization Cancellation (EC) principle [6] for auditory bandpass-filters below 1.5 kHz and better ear listening for filters above 1.5 kHz. EC processing involves removing interaural level and time differences from the ear signals for each filter before simultaneously subtracting and adding them. In [2] the Speech-to-Reverberation Modulation energy Ratio (SRMR) measure [7] had been used to determine if subtraction or addition was more beneficial in terms of enhancing speech and the better signal was selected for signal reconstruction. For computational efficiency a more low-level approach compared to the SRMR has been implemented, which compares the energy of one modulation bandpass-filter centered around modulation frequencies typical for speech to another filter centered around modulation frequencies, which are caused by noise or reverberation. The front-end then outputs a monaural signal to the back-end.

The end-to-end trained version of LEAP was trained using the outputs of the original implementation as targets to mimic the behavior. The end-to-end version takes mel-spectrograms as inputs and consists of several consecutive one dimensional convolution steps in time and frequency direction. Training data was generated by using multiple freely available everyday noise and music as well as german speech corpora. Data augmentation of different types like, e.g., applying reverberation was applied to make the model robust against common disturbing factors.

Parameter Profiles

The profile approach from [5] was implemented for assessing subjective LE in real-time with some changes. While they generated profiles for signal distortion, which contained gradual slope-like changes, in this study only abrupt changes were used. This was done to estimate response times more reliably. Also, instead of adapting distortion, SNR and reverberation T_{60} were used separately as well as simultaneously leading to three distinct profiles. Profiles adapting just SNR and T_{60} are further referred to as "SNR only" and "'T_{60} only" respectively, while the profile adapting those parameters simultaneously is referred to as "SNR + T_{60}". To investigate the effect of adapting parameters separately, if only SNR was adapted, no reverberation was used and if only T_{60} was adapted, no noise was used. SNR values were applied by amplifying speech and attenuating noise by half the SNR respectively, where for an SNR value of 0 dB both signal types were played back at a level of 65 dB SPL. This was done to avoid very high overall levels, which could cause

increased subjective LE due to high loudness. For the abrupt changes SNR values were drawn at random from a distribution of -12 to +12 dB in steps of 3 dB while T_{60} values were chosen to be 0, 1, 3 or 5 s at random. Each profile was of 300 s length and contained 30 sections, where the respective parameters stayed constant. This would allow for choosing a constant section length of 10 s, however participants could get used to a constant interval between sections and anticipate when a parameter change will happen. Therefore, section lengths were chosen to be either 7, 10 or 13 s, while each of those lengths were making up a third of all sections and the order of occurrence was randomized. Furthermore, a gain roving was applied to each section in a range of -3 to +3 dB in steps of 1 dB, which was applied to the whole stimulus gain. This was done to avoid measuring subjective LE at a constant loudness.

In order to investigate the model's capability to predict a binaural benefit due to spatial release of masking profiles were presented in two spatial conditions, which were simulated using the Toolbox for Acoustic Scene Creation and Rendering (TASCAR) [8]. For the first condition speech and noise were located in front of the listener, which will be referred to as the S_0N_0 condition. For the second condition speech and noise were lateralized each by 45° in opposite directions, which will be referred to as the $S_{-45}N_{+45}$ condition. When convolving the room impulse responses with the audio of a section, the previous four seconds of audio were included and discarded afterwards, so that reverberation did not have to build up inside the sections. This lead to a total of 6 profile presentations. Additionally, a training profile was generated, where for each section any out of the 6 profile situations appeared. The receiver was placed at a height of 1.80 m inside a 30 m × 16 m × 9.1 m room and slightly moved from the room center by 5 m and 3 m in x- and y-direction to allow for as many different reflection paths as possible. For speech a German narrated tale was used, which was adapted by [9] so that speech pauses were limited to 500 ms and the whole audio book cut into segments of 10 min each. The segments were further cut into segments of 5 min, which is the same length as the profiles and will further be referred to as chapters. Additionally, questionnaires from [9] were adapted to be usable for each chapter in order to check, if participants were paying attention to the story contents throughout the hearing experiment. In case noise was present, ICRA1 noise was used [10]. To keep the context of the story the chapters were always played back in chronological order, while the order of profiles was randomized for each participant. The training profile was applied to another German tale used and adapted by [9], so that each participant heard the same training stimulus.

Measurements

A slider interface for a smartphone was implemented. The slider displayed categories linked to hidden Effort Scale Categorical Units (ESCU) [11]. Categories from "no effort" to "extreme effort" plus an additional category "noise only" were displayed with corresponding values of 1 to 13 plus 14 ESCU. As an additional visual cue the background slider was changing depending on the slider position ranging from green to red.

The hearing experiment was conducted inside a soundproof booth and stimuli were presented using Sennheiser HD650 headphones. The headphones were calibrated with Brüel & Kjær hardware consisting of a Type 4153 artificial ear, a Type 2669 preamplifier and a Type 2610 measuring amplifier calibrated with a Type 4231 sound calibrator playing back a 1 kHz sine at 94 dB SPL.

Participants needed to fulfill several requirements. First, they had to be normal-hearing, meaning that no hearing loss greater than 20 dB HL had to be measured according to PTA4, which was checked before the experiment. Also, they had to have no reported color blindness in order to benefit from the visual color changing cue from the slider. Lastly, participants had to score at least 50 % in the questionnaire. In total, 19 participants took part in the experiment due to one participant being excluded, of which just 3 reported to have taken part in at least one experiment, where LE was assessed.

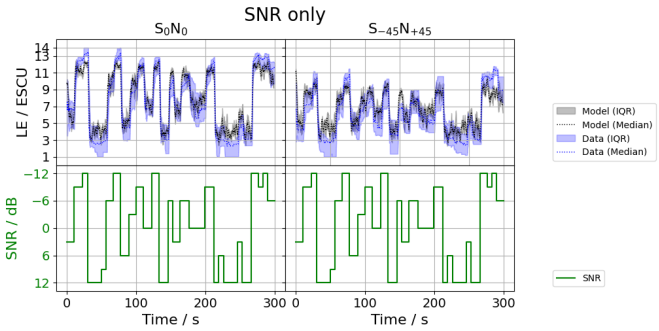


Figure 1: Median subjective LE of participants and LE predicted by the model in ESCU for both spatial conditions S_0N_0 and $S_{-45}N_{+45}$ (upper panels) and the presented "SNR only" profile plotted with inverted y-axis (lower panels). The blue patch in the LE plots show the Inter Quartile Range (IQR).

Results

The time course of subjective LE for both spatial conditions for the profile adapting only the SNR can be seen in Figure 1. It shows the median subjective LE for participants and LE predicted by the model over the course of time as well as the Inter Quartile Range (IQR). The model predictions were calculated by presenting all six combinations of one profile with the six chapters and calculating the median and IQR for those time courses. The y-axis of the SNR profile was inverted so that high values correspond to low LE and vice versa.

As can be seen in Figure 1 the median subjective LE of participants and LE predicted by the model follow the course of the profile with a small time lag regardless of the spatial condition. Also the subjective LE and IQR of participants and LE predicted by the model are very similar, while the overall accuracy is lower if speech and noise are spatially separated. Additionally, the overall subjective LE for the $S_{-45}N_{+45}$ condition is lower compared to the S_0N_0 condition. The decrease in subjective LE depends on the SNR, where in the most extreme cases in sections with SNR values of -12 dB the difference is up

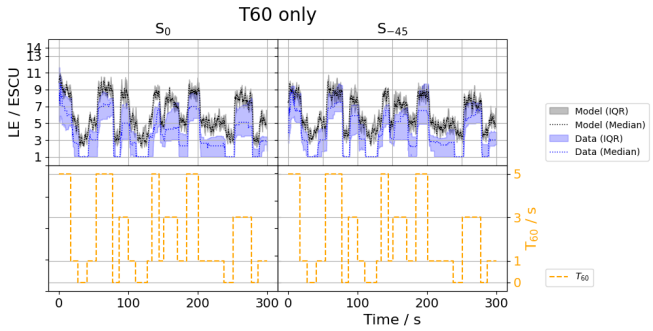


Figure 2: Median subjective LE of participants and LE predicted by the model in ESCU for both spatial conditions S_0N_0 and $S_{-45}N_{+45}$ (upper panels) and the presented "T_{60} only" profile (lower panels). The blue patch in the LE plots show the IQR.

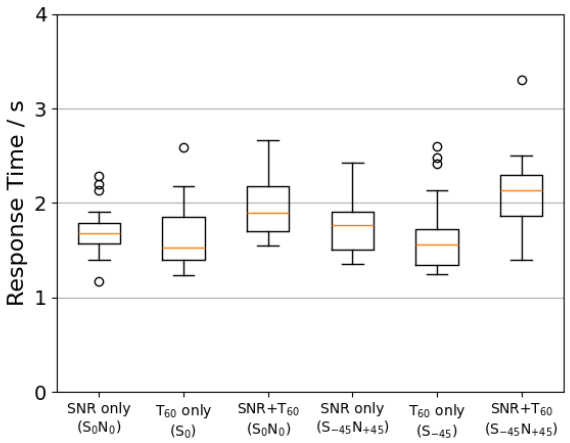


Figure 3: Response times of participants in s for each of the six profiles.

to 3 ESCU, while for sections with SNR values of +12 dB the difference drops below 1 ESCU or is not observable anymore.

For the "T_{60} only" profile the time courses can be seen in Figure 2. Since no noise was present the spatial conditions are labeled as S_0 and S_{-45} indicating that the only difference was the lateralization of the speech source by -45°. In contrast to the "SNR only" profile in both spatial conditions subjective LE by participants did not exceed 7 ESCU even for a T_{60} of 5 s except for the beginning. The lateralization of the speech source did not lead to large changes in subjective LE like for the "SNR only" profile, while in some sections subjective LE even increased slightly. The LE predicted by the model exceeds subjective LE by participants throughout the stimuli by up to almost 4 ESCU in the most extreme cases.

For estimating the response times of participants for each profile, the cross correlation between the participants subjective LE time courses and the course of the respective profile was calculated. The point in time of the cross correlation peak of each participant was then interpreted as the average response time. The response times for each profile can be seen as a box plot in Figure 3. Overall, the median response times are mostly just below 2 s, while there's a slight tendency towards higher response times for the profiles, where SNR and T_{60} are adapted simultaneously. However, applying a Kruskal-Wallis test due to non-normally distributed data revealed that no response time distribution for any profile differed significantly from the others.

Discussion

The continuously rated subjective LE clearly follows the course of the profiles for the whole duration of the experiment, while subjective LE increases for decreasing SNR and increasing T_{60} respectively. As can be seen in the time courses of subjective LE for participants for the "SNR only" profile, the effect of spatial release of masking is observable and decreases with increasing SNR. The same can be seen for the model predictions. When comparing this to the findings from [1], where subjective LE was assessed using a static procedure playing back sentences of speech in speech-shaped noise at different SNRs, the effect appears to be less extreme here. Compared to their S_0N_{-100} condition, which would be closest to the angular distance of the sources here of 90°, spatially separating the sources lead to a decrease of up to 6 ESCU for an SNR of -10 dB for participants as well as for the evaluated models. However, these deviations could be the consequence of the differences in the methods.

The extreme reverberation times used here without interfering noise did not lead to extreme subjective LE as intended. However, when comparing these findings to the ones from [1] for sentence tests, where the most comparable condition was a setting with a Direct-to-Reverberant Ratio (DRR) of -17 dB with noise at an SNR of +10 dB and the IQR of subjective LE for participants reached up to 7 ESCU, the results appear to be comparable. In case of added noise for the "SNR + T_{60} profiles, of which the results are not shown here, an increase comparable to [1] is observable for the setting with a DRR of -17 dB.

Regarding the estimated response times the average response time is about 2 s, while there's no significant deviation for any profile type. Compared to the delay of 1 s for continuously assessed speech quality in [5] the response time in this study appears to be quite high. One possible explanation could be that changes in speech quality were more noticeable or easier to rate for participants. Another reason might have been that in [5] participants were recruited from the laboratory and might have had more experience in experiments, where speech quality is assessed. Their participants also did not additionally have to pay attention to the spoken words, which could have increased the overall total mental load in this study. Also, the use of fewer labels on the slider in their study could have made it easier for participants to decide on the rating.

The average response time of 2 s suggests that a real-time model could get away with making predictions with delays in the order of magnitude of seconds to accurately model the temporal lag of participants. However, the fact that participants have to coordinate and move their finger on the slider, which takes up some of the response

time, has to be considered. To estimate the exact delay for perceiving LE more sophisticated methods would have to be applied, which could incorporate objective measurements. However, then the question arises if such methods have the temporal resolution and can provide instantaneous LE or also introduce delays. The method presented in this study has the advantage that it is easily set up and could in theory be used for in-the-field studies for experiments with focus on ecological validity.

Conclusion

A hearing experiment assessing continuous subjective LE of participants has been conducted in order to evaluate a real-time model that predicts LE. Participants and the model could follow changes of SNR or T_{60} very reliably, while the model systematically overestimates the influence of reverberation. Further model adaptions could address this problem, while for now it is unclear if the bias is caused by the front-end or the back-end. Furthermore, model predictions may be smoothed to match the response time of participants of 2 s. When using this model in a hearing aid to select the best algorithm in the moment, adaption times could be even more relaxed though, which could be investigated in another experiment.

The method introduced in this study for assessing continuous subjective LE could even be extended to work outside the laboratory due to the easy and lightweight setup, which could be a next step towards ecological validity. In a future experiment participants could be equipped with microphones at the ears to obtain signals for the model, while walking around with the slider application at points of interest regarding challenging listening situations in terms of perceived subjective LE.

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